Final Mission: Stop the Outbreak

What is the BEST strategy to prevent a future cholera outbreak?

The situation: *A cholera outbreak has started in a coastal city of 500,000 people. Vibrio cholerae is spreading through contaminated water. 200 people are already sick. The city has limited resources. You have been hired as a public health scientist. Your job: choose the BEST strategy to stop the outbreak and prevent the next one — then defend your choice with a Claim, Evidence, and Reasoning (CER).*

PART A — Review Your Options

Study the five strategies below. Use what you've learned across the whole unit to evaluate each one.

- **Clean Water, Sanitation & Hygiene**

  Fix the water supply, build sewage systems, distribute soap and oral rehydration solution (water + salt + sugar). John Snow used this approach to stop the 1854 London cholera outbreak by removing a contaminated pump handle.

- **Smarter Antibiotic Use**

  Treat infected patients with antibiotics. Educate the public to only take antibiotics when prescribed and to always finish the full course.

- **Phage Therapy**

  Use bacteriophages — viruses that only attack bacteria — to target Vibrio cholerae directly. This avoids creating antibiotic resistance and can target specific bacteria.

- **Protect the Microbiome**

  Distribute probiotics to residents to strengthen gut microbiomes and reduce vulnerability. A diverse gut microbiome is better at resisting pathogens.

- **Block Bacteria's Signals**

  Use quorum sensing inhibitors to prevent Vibrio cholerae from coordinating an attack. Still experimental but could prevent biofilm formation and reduce virulence.

PART B — Evaluate Each Strategy

Rate each strategy 1–5 (5 = strongest) for each criterion. Then add up your scores.

Strategy	Stops current outbreak? (1–5)	Prevents future outbreaks? (1–5)	Avoids resistance? (1–5)	Total
■ Clean water				
■ Antibiotics				
■ Phage therapy				
■ Microbiome				
■ Quorum sensing				

PART C — Your CER — Defend Your Strategy

Choose the strategy you rated highest (or argue for a combination). Write a full CER response.

CLAIM — State which strategy is best in one clear sentence.

EVIDENCE — Use at least THREE specific pieces of evidence from the unit. Reference specific stops, data, or examples.

REASONING — Explain WHY your evidence supports your claim. Connect the biology to the real-world outcome.

COUNTERARGUMENT — What is the strongest argument AGAINST your strategy? How would you respond to it?

PART D Rubric

Category	4 — Exceeds	3 — Meets	2 — Approaching	1 — Beginning
Claim	Clear, specific, and arguable	Clear and arguable	Vague or incomplete	Missing or unclear
Evidence	3+ specific examples from unit with detail	3 examples, some detail	1–2 examples, little detail	No specific evidence
Reasoning	Explicitly connects evidence to claim with biology concepts	Connects evidence to claim	Partial connection	No connection
Counterargument	Names strong objection and rebuts with evidence	Names objection and responds	Names objection only	Missing

■ **Unit Reflection** We started with the question: *Why are bacterial diseases coming back, and how can we stop the next outbreak?* In 3–4 sentences, answer this question using evidence from across the whole unit. Include at least 4 vocabulary words.

Adapted for 7th grade from New Visions for Public Schools, Humans vs. Bacteria (Biology Unit 2), CC BY-NC-SA 4.0. Unit page: yscabarney.github.io/MS_SCI_ADV_26/humans-vs-bacteria.html